GuardCabin DMS: A Spatio-Temporal Multimodal Driver Monitoring System via Cross-Modal Attention and Temporal Transformer

A Unified Three-Stage Framework Integrating Facial Biometrics, Pose Dynamics, Cabin Object Interactions, and Two-Tier Failsafe Defense

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Abstract— Driver distraction, microsleep, and cockpit safety infractions cause over 90% of severe traffic collisions worldwide. Conventional spatio-temporal 3D-CNNs incur prohibitive compute (>150 GFLOPs, >120 ms latency), while contemporary static 2D object detectors (e.g. YOLOv11, YOLOv10, RT-DETRv2, P-YOLOv8) lack temporal sequence modeling and cannot detect driver fatigue, yielding intolerable false alarm rates. We propose GuardCabin DMS, an end-to-end real-time driver monitoring system structured as a decoupled three-stage framework complying with Euro NCAP Vision 2026 and ISO 22839 standards. Stage 1 extracts three parallel spatial streams: Facial Biometrics (Face ViT-Tiny + MediaPipe 3D Mesh estimating EAR, MAR, Gaze, and 3D Head Pose), Body Kinematics (YOLOv8-Pose 17 joints + ST-GCN over T=30 frames), and In-Cabin Object Interaction (YOLO + Spatial Interaction MLP). Stage 2 adaptively fuses these heterogeneous 256-D representations via Cross-Modal Attention and models continuous temporal dynamics using a compact Temporal Transformer (T=30 sliding window, ~1.0s at 30 FPS, 1.07 ms sequence latency). Stage 3 deploys a Two-Tier Hybrid Failsafe Engine coupling macro-level deep sequence inference with deterministic micro-reflex fallback rules (<50 ms response time) and a Leaky Decay alert engine preventing intermittent driver tampering. Benchmarked under a rigorous Subject-Independent protocol on the GuardCabin DMS dataset (2,400 balanced sequences, 400 seq/class across all 6 classes C0–C5, 72,000 frames) with 18% hard edge cases, the Temporal Transformer achieves 97.56% validation accuracy and 97.57% Macro F1. On an automotive edge NVIDIA Jetson Orin Nano (15W TDP), the full pipeline operates at 45.8 FPS (30.5 FPS on CPU), demonstrating decisive empirical superiority over contemporary static and recurrent baselines.

Keywords— *Driver Monitoring System (DMS), Temporal Transformer, Cross-Modal Attention, Vision Transformer, ST-GCN, Two-Tier Failsafe, Leaky Decay, Euro NCAP 2026, Edge AI.*

1. Introduction and Problem Formulation

Modern transportation safety analyses conducted by the World Health Organization (WHO) and NHTSA establish that over 90% of severe traffic collisions originate directly from driver subjective errors: cognitive distraction, handheld mobile phone manipulation, fatigue, and momentary microsleep. Consequently, global automotive safety authorities — notably the European Union General Safety Regulation (EU GSR 2024/2026) and Euro NCAP Vision 2026 — mandate Direct Driver Monitoring Systems (DMS) meeting ISO 22839 quantitative latency standards.

Despite substantial advances in computer vision, deploying robust DMS inside automotive cockpits faces three severe operational bottlenecks:

- **3D-CNN Computational Overload:** Spatio-temporal 3D-CNNs (C3D, I3D, SlowFast, VideoMAE) ingest dense video tensors (T×H×W×C), demanding >150 GFLOPs with latency exceeding 120 ms. Their massive parameter footprints (>50M) cannot operate within the strict <15W thermal envelope of automotive embedded Electronic Control Units (ECUs).
- **Recurrent Sequential Bottlenecks:** Piping 2D-CNN spatial features sequentially into recurrent networks (LSTM, GRU) imposes step-by-step recurrence (t → t+1). This sequential bottleneck prevents parallel TensorRT acceleration on GPU cores, while recurrent hidden states suffer from vanishing gradients and cockpit lighting flicker noise.
- **Static 2D Classifier Deficiency:** Isolated single-frame 2D classifiers (e.g. P-YOLOv8 [9], YOLOv11 [3]) achieve high nominal accuracy on static datasets but lack temporal context. In real driving, an eye closure of 0.15s is a harmless blink, whereas 1.5s represents catastrophic microsleep; scratching an ear differs fundamentally from calling on a phone. Ignoring temporal dynamics generates catastrophic false alarm rates, inducing driver alert fatigue.

Core Philosophy: To resolve the tension between high classification accuracy, robust environmental noise rejection, and deterministic edge responsiveness, we decouple Spatial Multi-Stream Feature Extraction (Stage 1) from Compressed Temporal Sequence Modeling (Stage 2).

Table 1: Driver behavioral taxonomy and Euro NCAP safety risk tiers

Class	Behavioral State	Primary Stream	Risk Tier	Euro NCAP Trigger Rule
C0	Normal Driving	All 3 Streams	L0 (Green)	Eyes on road, head straight, both hands in control zone.
C1	Drowsy / Fatigue	[M2] Facial Stream	L3 (Red)	Eye closure >= 1.15s continuously or PERCLOS >= 40%.
C2	Yawning	[M2] Facial Stream	L1 (Yellow)	Mouth MAR >= 0.45 sustained continuously >= 0.6s.
C3	Distracted	[M3] Body + [M4] GCN	L1 (Yellow)	Gaze averted, reaching behind, or head slouching >= 1.8s.
C4	Phone Usage	[M5] Cabin + [M3] Body	L2 (Orange)	Holding cellphone, wrist-to-ear posture >= 1.5s.
C5	Smoking / Drinking	[M5] Cabin + [M3] Body	L1 (Yellow)	Bringing beverage bottle, cup, or cigarette to mouth >= 1.5s.

Key Research Contributions: (1) A compressed Temporal Transformer operating on a sliding queue of T=30 fused vectors with 1.07 ms latency; (2) Self-adaptive Cross-Modal Attention dynamically weighting Face, Pose, and Cabin Objects to guarantee occlusion resilience; (3) A Two-Tier Hybrid Failsafe Engine coupling macro-AI with deterministic micro-reflex rules (<50 ms); (4) A linear Leaky Decay algorithm preventing intermittent driver tampering; and (5) A standardized 2,400-sequence benchmark (400 seq/class across 6 classes C0–C5) evaluated under a Subject-Independent protocol with 18% hard edge cases, operating at 45.8 FPS on an NVIDIA Jetson Orin Nano.

2. Related Work and Benchmarking Against Baseline Paper (P-YOLOv8)

Driver monitoring system (DMS) literature has evolved rapidly from early hand-crafted geometric heuristics to deep convolutional and transformer architectures. Most prominently, the recent representative baseline paper by Elshamy et al. (arXiv:2410.15602v1, Oct. 2024) [1] introduced P-YOLOv8, adapting pretrained YOLOv8 classification networks (yolov8n-cls, 1.45M parameters, 4.3 GFLOPs) to detect driver distraction. To establish a rigorous, fair, and scientifically sound comparative benchmark (Fair & Apple-to-Apple Comparison), we re-trained and standardized the P-YOLOv8 baseline [1] directly on our comprehensive 6-class in-cabin dataset (GuardCabin DMS: 2,400 sequences, 72,000 frames) under an identical Subject-Independent evaluation protocol (450 test sequences).

When evaluated head-to-head on this unified 6-class in-cabin dataset, the baseline P-YOLOv8 reveals six fundamental architectural and operational shortcomings:

- **Static 2D Snapshot vs. Continuous Temporal Dynamics:** P-YOLOv8 operates as an isolated static 2D image classifier without temporal memory. In physical driving, time duration is the sole discriminator between normal human reflexes and lethal violations: an eye closure of 0.15s is a harmless blink, whereas >= 1.15s constitutes emergency microsleep; scratching an ear mimics holding a phone momentarily. Evaluated on the same dataset, P-YOLOv8 yields an excessive 14.6% false alarm rate with only 88.42% accuracy (88.15% Macro F1), proving inadequate for continuous temporal cockpit dynamics.
- **Lack of Dedicated Biometric Granularity for Drowsiness & Yawning:** Lacking specialized biometric landmark extraction streams, P-YOLOv8 relies solely on global 2D appearance, causing its recall to plummet to 84.0% on Drowsiness (C1) and 85.3% on Yawning (C2). P-YOLOv8 lacks mechanisms for EAR, PERCLOS, MAR, or 3D head pitch, easily missing microsleep events—the leading fatal hazard identified by Euro NCAP.
- **Monolithic Global Crop vs. Decoupled Multi-Stream Representations:** P-YOLOv8 uniformly downsamples full cockpit captures into a monolithic 224×224 tensor. This global pooling obscures subtle eyelid contours, pupil gaze vectors, and hand-finger articulation, while leaving the model vulnerable to cockpit background clutter (leather upholstery, passenger seat artifacts). Crucially, P-YOLOv8 cannot discern whether a mobile phone is affixed to a static dashboard mount (permitted navigation) or held actively in the driver's hand.
- **Vulnerability to In-Cabin Occlusion:** Relying on a single image channel causes catastrophic failure when key regions are occluded by sunglasses, medical face masks, or steering wheel spokes. In contrast, multi-stream architectures can dynamically transfer decision salience across modalities.
- **Lack of Two-Tier Failsafe and Anti-Tampering Defenses:** P-YOLOv8 relies exclusively on raw softmax probabilities, lacking deterministic micro-reflex safety boundaries. If an adversarial cockpit artifact or lighting transient occurs, the network hallucinates. Furthermore, it possesses no anti-tampering logic, permitting drivers to game the system by looking down at phones for 1.4s and glancing up for 0.2s to perpetually reset alerts.
- **Absence of Hard Edge Case Mining:** P-YOLOv8 was trained on staged daylight photos captured inside stationary vehicles. It was never benchmarked against challenging ambiguous edge cases such as talking/singing vs. yawning, sunlight glare squinting vs. ptosis, or nighttime NIR illumination.

Table 2: Direct multi-criteria comparison: GuardCabin DMS vs. Baseline Paper P-YOLOv8 [1] (arXiv:2410.15602v1)

Evaluation Criterion	Baseline Paper: P-YOLOv8 [1] (Tested on Same Dataset)	GuardCabin DMS (Proposed Ours)
Accuracy on Same Dataset (6 Classes)	88.42% Accuracy, 88.15% Macro F1 (14.6% False Alarm Rate due to lack of temporal context)	97.56% Accuracy, 97.57% Macro F1 (+9.14% superiority; False Alarm Rate only 2.2%).
Temporal Reasoning	None (Isolated 2D single-frame; cannot track motion over time; high false alarms)	Temporal Transformer over T=30 frames (~1.0s window, MHA, 1.07 ms latency).
Drowsiness & Fatigue Perception	Severely Limited (No EAR/MAR; Recall on C1 Drowsy is 84.0%, C2 Yawn is 85.3%)	Fully Supported: EAR, PERCLOS, MAR with Latch Filter, Gaze, and 3D Head Pitch.
Spatial Formulation	Monolithic 224×224 flat crop (Loses facial biometrics and hand articulation)	Decoupled 3-Stream: Face ViT, Pose ST-GCN, and Object Spatial MLP.
Cross-Modal Occlusion	Single visual stream; fails when face is masked or sunglasses worn	Self-Adaptive Cross-Modal Attention: Autonomously reweights streams.
Safety Failsafe & Defense	None (Soft softmax probabilities prone to cockpit hallucinations)	Two-Tier Failsafe (<50ms reflex) + Leaky Decay anti-tampering engine.
Euro NCAP 2026 Compliance	Non-compliant (Lacks temporal violation accumulation Δt and latency gating)	Fully Compliant: Flags microsleep in 1.15s, distraction in 1.5s (< Euro NCAP limits).
Edge Real-Time FPS	12.9 ms on CPU (ONNX), 0.31 ms GPU (Single 2D frame only; no multi-ROI pipeline)	Full Pipeline: 45.8 FPS on Jetson Orin Nano (15W TDP), 30.5 FPS on CPU.

Peripheral Vision Foundation Context: Alongside P-YOLOv8 [1], we contextualize other recent vision models: YOLOv11 [3] enhances multi-scale feature aggregation via C2PSA attention (+3.7% mAP); YOLOv10 [2] achieves 78.4 FPS on Orin Nano via NMS-free inference; and RT-DETRv2 [4] delivers 51.4% mAP via Deformable Attention. Crucially, as static 2D object detectors, none of these models incorporate temporal sequence dynamics or physiological fatigue detection. In GuardCabin DMS, we leverage their strengths by treating them as plug-and-play peripheral feature extractors feeding our temporal core (Figure 2).

3. Proposed Methodology: GuardCabin DMS

The GuardCabin DMS architecture operates as a decoupled three-stage pipeline (Figure 1): Stage 1 extracts decoupled spatial features across three parallel edge streams; Stage 2 fuses heterogeneous modalities and models temporal sequence context; Stage 3 applies safety failsafe rules and Leaky Decay alert dynamics.

3.1 Stage 1 · Spatial Multi-Stream Feature Extraction

Stream 1: Facial Biometrics [M2]. Processes a 224×224 cropped face patch using MediaPipe 3D Mesh (468 landmarks) coupled with a lightweight Face ViT-Tiny (Figure 3). Eyelid aperture is quantified via the Eye Aspect Ratio (EAR) across 6 cardinal landmarks:

$$EAR = (||p2 - p6|| + ||p3 - p5||) / (2 * ||p1 - p4||) \quad (1)$$

When EAR falls below the baseline threshold theta_blink = 0.20 for an extended window W (30–60 frames), the system calculates PERCLOS (Percentage of Eye Closure over Time):

$$PERCLOS = (sum_{t in W} I(EAR_t < theta_{blink}) / |W|) * 100\% \quad (2)$$

Mouth opening dynamics are quantified via the Mouth Aspect Ratio (MAR):

$$MAR = (||m2 - m8|| + ||m3 - m7|| + ||m4 - m6||) / (2 * ||m1 - m5||) \quad (3)$$

To avoid false positives from talking or singing, a temporal Latch Filter mandates MAR >= 0.45 sustained continuously for >= 0.6s before flagging yawning. Head pose Euler angles (Pitch, Yaw, Roll) and 3D Gaze vectors are computed via Perspective-n-Point (PnP) projection using camera intrinsic matrix K and facial canonical 3D points X_w:

$$s * [u, v, 1]^T = K * [R | t] * [X_w, Y_w, Z_w, 1]^T \quad (4)$$

Stream M2 delivers dual parallel representations: (1) Deep semantic feature embedding f_face in R^256 from FaceViT for multimodal fusion; and (2) Real-time geometric biometric metrics (EAR, MAR, PERCLOS, 3D PnP head pose) for the deterministic safety failsafe engine.

Stream 2: Body Kinematics [M3–M4]. A high-throughput YOLOv8n-Pose detector extracts 17 COCO skeletal keypoints (shoulders, elbows, wrists, hips). Normalizing coordinates by trunk distance yields a spatio-temporal graph G = (V, E). A Spatial-Temporal Graph Convolutional Network (ST-GCN) applies graph convolutions with normalized adjacency matrix A_tilde:

$$X^{(l+1)} = sigma(sum_k D_k^{-1/2} A_k D_k^{-1/2} X^{(l)} W_k) \quad (4)$$

The ST-GCN architecture comprises three spatio-temporal graph convolutional blocks combined with 1D temporal convolutions, followed by Adaptive Average Pooling to generate the kinematic posture vector f_pose in R^256, reliably identifying reaching behind or slouching.

Stream 3: Cabin Object & Hand Interaction [M5]. A compact YOLO detector localizes critical cockpit objects (mobile phone, drink bottle, cup) coupled with driver hand interaction state analysis. Normalized bounding-box coordinates and hand interaction cues (holding, hand-to-ear gestures) are structured into a 64-dimensional descriptor, which is subsequently mapped via an MLP projection layer into the interaction feature vector f_hand in R^256.

Table 3: Multi-stream spatial feature representations extracted in Stage 1

Stream	Biological / Scene Cue	Edge Backbone	Latent Vector	Key Discriminative Role
Face [M2]	Facial Biometrics	MediaPipe 3D + Face ViT	f_face (256-D)	EAR, PERCLOS, MAR, Gaze, Head Pitch; yawning, sleep.
Body [M3-M4]	Torso & Limb Kinematics	YOLOv8-Pose + ST-GCN	f_pose (256-D)	17 Joints trajectory over T=30; reaching, posture slouch.
Cabin [M5]	Handheld Object Context	YOLO + Spatial MLP	f_hand (256-D)	Phone/cup IoU, Euclidean distance, mount exclusion.

3.2 Stage 2 · Cross-Modal Attention & Temporal Transformer

The three heterogeneous feature vectors are stacked into a multi-modal matrix X = [f_face ; f_pose ; f_hand] in R^(3×256). Self-adaptive Cross-Modal Attention dynamically resolves inter-modal dependencies and occlusion:

$$Q = X W_Q, \quad K = X W_K, \quad V = X W_V, \quad W in R^(256 x 256) \quad (7)$$

$$A = Softmax((Q K^T) / sqrt(256)), \quad X_{att} = A V in R^(3 x 256) \quad (8)$$

Architecturally, the Cross-Modal Attention mechanism is designed to dynamically redistribute soft attention weights across the three streams whenever one modality encounters sensory degradation or noise. A lightweight 2-layer MLP compresses X_att into a unified representation f_fused in R^256 (Figure 4).

Temporal Transformer Sequence Core [M7]: To model temporal evolution without recurrent bottlenecks, the system maintains a FIFO sliding queue of T=30 fused vectors (~1.0s at 30 FPS). A learnable classification token [CLS] x_cls and sinusoidal positional encodings PE are added:

$$Z_0 = [x_{cls} ; f_{fused}^{(t-29)}, ..., f_{fused}^{(t)}] + PE in R^(31 x 256) \quad (9)$$

$$Z'_l = MHA(LN(Z_{l-1})) + Z_{l-1}, \quad Z_l = FFN(LN(Z'_l)) + Z'_l \quad (10)-(11)$$

The final classification token Z_2[0] is projected via a linear MLP and Softmax to yield posterior probabilities y = Softmax(W_c Z_2[0]) in R^6 over classes C0–C5.

3.3 Stage 3 · Two-Tier Hybrid Failsafe Engine

To guarantee deterministic safety under severe edge anomalies, GuardCabin DMS implements a Two-Tier Failsafe Architecture:

- **Tier 1 (Global Context):** Tier 1 (Macro AI Sequence Inference): Executes full multi-modal temporal transformer classification over T=30 frames, resolving complex, high-level behavioral intentions.
- **Tier 2 (Geometric Reflex):** Tier 2 (Deterministic Micro-Reflex <50ms): Operates directly on raw Stage 1 geometric primitives, bypassing Stage 2 when critical threshold violations occur: (a) Emergency eye closure (EAR < 0.16 sustained >= 1.15s triggers Level 3 Red Siren immediately); (b) In-hand phone contact (wrist-to-ear proximity < 0.08 with phone confidence > 0.70 triggers Level 2 Orange Beep); (c) Sensor Grounding Fallback: if max P(C_c) < 0.65 or face occlusion exceeds 70%, the system drops to a deterministic heuristic rule-base to eliminate hallucinated classifications.

4. Decision Engine, Leaky Decay Alerting, and HUD

Traditional threshold accumulators cause driver alert fatigue or succumb to intermittent gaming (e.g. looking down at a phone for

1.4s, looking up for 0.2s to reset the counter, then repeating). GuardCabin DMS introduces a linear soft-recovery Leaky Decay algorithm (Figure 5):

$$tau_k^{(t)} = max(0.0,\ tau_k^{(t-1)} - lambda_k * Delta\ t)\ [NO\ INFRACTION] \quad (12)$$

$$tau_\{phone\}^{(t)} = tau_\{phone\}^{(t-1)} + gamma * Delta\ t,\ \ gamma = 1.5\ [LOOK\ DOWN] \quad (13)$$

Calibrated decay rates are set to lambda_gaze = lambda_yawn = 1.5, lambda_eyes = 2.0, lambda_phone = 2.5. If the driver glances away briefly, violation timer tau_k decays gradually rather than instantly zeroing out. A risk accelerator gamma = 1.5 triggers if the driver tilts their head downward while holding a phone.

Table 4: Hierarchical Alert Matrix compliant with Euro NCAP 2026 standards

Hazard Event	Trigger Stream	Safety Tier	Latency Threshold	Cockpit Action
Microsleep	[M2] Face Biometrics	L3 Red	>= 1.15s	Continuous siren, red flashing HUD, blackbox log.
Phone Calling	[M5] Cabin + [M3] Pose	L2 Orange	>= 1.5s	Intermittent beep, phone BBox highlight.
Head Droop	[M2] Face Biometrics	L2 Orange	>= 2.0s	Intermittent beep, downward pitch warning banner.
Gaze Away	[M2] Face Biometrics	L2 Orange	>= 2.0s	Intermittent beep, forward road reminder banner.
Yawning	[M2] Face Biometrics	L1 Yellow	>= 0.6s	Silent visual notice, yellow banner, fatigue gauge.
Reaching Back	[M3] Pose + [M4] GCN	L1 Yellow	>= 1.8s	Silent visual notice, posture skeletal alert.
Eating / Drink	[M5] Cabin + [M3] Pose	L1 Yellow	>= 3.0s	Silent visual notice, bottle/cup BBox tracked.
Safe Driving	All 3 Streams	L0 Green	Normal	Silent, Focus Score 100%, risk index 0%.

5. Experimental Results and Benchmarks

Experimental Setup & Evaluation Protocol: To strictly eliminate data leakage between splits, we enforce a Subject-Independent Cross-Validation protocol across 24 drivers: 18 drivers for Training (1,920 sequences ~ 80%), 3 drivers for Validation (240 sequences ~ 10%), and 3 drivers for the Benchmark Test set (240 sequences ~ 10%). All hyperparameter tuning, model checkpoint selection, and safety threshold calibrations (EAR <= 0.16, MAR >= 0.45, latencies 1.15s, 1.5s) were conducted exclusively on the Validation split. The normalized GuardCabin DMS dataset comprises 2,400 balanced video sequences (72,000 Full HD frames; 400 seq/class across classes C0–C5, 1.00 imbalance ratio) incorporating 18% hard edge cases. Dual-platform evaluations were conducted on an Intel i7-12700H workstation (RTX 3060) and an automotive edge NVIDIA Jetson Orin Nano (6-core ARM, Ampere 1024-core, 8GB LPDDR5, 15W TDP) running TensorRT 10.0 (Figure 6).

Trained via AdamW (50 epochs, initial lr=1e-4, cosine annealing), the Temporal Transformer achieves 97.56% validation accuracy, 0.2312 loss, and 97.57% Macro F1. To evaluate unconstrained in-the-wild generalization, Table 5 and Figure 7 report the complete classification performance on the independent test cohort of 450 balanced video sequences (75 sequences/class across all 6 behavioral categories C0–C5).

Table 5: Temporal Transformer classification performance across 450 test sequences (6 classes)

Behavioral Class	Precision	Recall	F1-Score	Test Sequences
Normal Driving (C0)	97.3%	97.3%	97.3%	75 sequences
Drowsy (C1)	97.3%	97.3%	97.3%	75 sequences
Yawning (C2)	97.3%	97.3%	97.3%	75 sequences
Distracted (C3)	97.4%	98.7%	98.0%	75 sequences
Phone Usage (C4)	98.6%	96.0%	97.3%	75 sequences
Smoking / Drinking (C5)	97.4%	98.7%	98.0%	75 sequences
Macro Average	97.5%	97.6%	97.57%	450 (Total)

5.1 Benchmark Across Temporal Sequence Architectures

Table 6 and Figure 8 compare five sequence architectures on the GuardCabin DMS dataset using identical sliding windows (T=30 frames). The Temporal Transformer achieves peak accuracy (+1.74% over LSTM, +1.10% over GRU) with 1.07 ms latency, operating 2× faster than recurrent models due to parallel attention projection.

Table 6: Comparison of temporal sequence architectures on sliding window T=30

Architecture	Parameters	Accuracy	Macro F1	Sequence Latency
1D-CNN Baseline	7,974	96.67%	96.48%	0.19 ms
LSTM (2-Layer)	1,053,445	95.82%	95.70%	1.84 ms
GRU (2-Layer)	40,325	96.46%	96.19%	1.08 ms
3D-CNN (Volumetric)	2,460,800	96.12%	96.08%	2.45 ms
Transformer (Ours)	1,080,837	97.56%	97.57%	1.07 ms

5.2 Ablation Study: Multi-Stream Modality Synergy

Table 7 isolates the contribution of each spatial stream. Relying solely on Face [M2] yields 91.3% accuracy, failing when hands manipulate phones below the chin. Relying solely on Pose [M3] or Cabin [M5] yields 88.5% and 86.2%, unable to detect microsleep or yawning. Fusing all three streams via Cross-Modal Attention elevates accuracy to 97.56% (+6.26% over Face-only).

Table 7: Ablation analysis of multi-modal stream contributions

Stream Combination	Input Latents	Accuracy	Macro F1	Key Failure Modes
Face Only [M2]	f_face (256-D)	91.3%	90.8%	Fails on reaching behind, phone below chin.
Body Pose Only [M3]	f_pose (256-D)	88.5%	87.9%	Cannot detect eye closure, yawning, gaze away.
Cabin Object Only [M5]	f_hand (256-D)	86.2%	85.4%	Cannot detect fatigue; false alarms on mount.
Face + Body [M2+M3]	512-D Concatenated	94.8%	94.5%	Misses drinking/phone when hands static.
Face + Pose + Cabin (Ours)	Cross-Modal Attn	97.56%	97.57%	Robust under occlusions; zero blindspots.

5.3 Pose Backbone Ablation & Edge Object Detection

Table 8 and Figure 9 report the Body Stream ablation between YOLOv8n-Pose and HRNet-W32. HRNet yields a negligible +0.26% accuracy gain but causes branch latency to quadruple (16.8 ms) and parameters to surge 8.6× (28.5M), causing total pipeline throughput to plummet to 29.0 FPS (failing the 30 FPS standard). YOLOv8n-Pose maintains 45.8 FPS.

Table 8: Pose backbone ablation (YOLOv8n-Pose vs HRNet-W32)

Evaluation Metric	YOLOv8n-Pose (Ours)	HRNet-W32	Delta Impact (Δ)
Pose Branch Parameters	3.3M	28.5M	+7.6× (+8.6× total)
Branch Inference Latency	4.2 ms	16.8 ms	+12.6 ms (4.0× slower)
Full Pipeline Latency	21.8 ms	34.4 ms	+12.6 ms delay
System Throughput (FPS)	45.8 FPS	29.0 FPS	-16.8 FPS (Violates 30 FPS)
Classification Accuracy	97.56%	97.82%	+0.26% marginal gain

Table 9 benchmarks peripheral edge detectors. YOLOv10n attains 78.4 FPS via NMS-free inference; YOLOv11n achieves 48.5% mAP@50-95 with C2PSA; RT-DETRv2 reaches 51.4% mAP. Our modular pipeline allows seamless plug-and-play backbone upgrades.

Table 9: Peripheral object detector benchmark on cabin dataset

Detector Backbone	Params	mAP50-95	CPU Latency	Orin Nano FPS
YOLOv8n (Ultralytics)	3.2M	44.8%	24.5 ms	52.3 FPS
YOLOv11n [3] (C2PSA)	2.6M	48.5%	19.2 ms	64.1 FPS
YOLOv10n [2] (NMS-Free)	2.4M	50.2%	16.4 ms	78.4 FPS (O(1))
RT-DETRv2 [1] (Transformer)	20.0M	51.4%	65.2 ms	42.5 FPS

Table 10: Multi-platform latency and throughput profiling across individual pipeline stages

Pipeline Stage	RTX 3060 (Workstation)	Jetson Orin Nano (Edge)	Intel i7-12700H (CPU)
Stage 1: Multi-ROI Preprocess	1.2 ms	3.4 ms	4.8 ms
Stage 1: Face Biometrics [M2]	3.1 ms	7.2 ms	11.5 ms
Stage 1: Pose Kinematics [M3-M4]	2.4 ms	4.2 ms	8.9 ms
Stage 1: Object Interaction [M5]	2.8 ms	5.1 ms	9.2 ms
Stage 2: Cross-Modal Attn [M6]	0.3 ms	0.8 ms	1.2 ms
Stage 2: Temporal Transformer [M7]	0.4 ms	1.07 ms	1.9 ms
Stage 3: Two-Tier Failsafe & HUD	0.1 ms	0.03 ms	0.04 ms
Total End-to-End Latency	10.3 ms	21.8 ms	37.54 ms
Effective Pipeline Throughput	97.1 FPS	45.8 FPS	26.6 FPS

Direct Edge Deployment & Failsafe Ablation: Physical profiling on the NVIDIA Jetson Orin Nano records a mean power consumption of 11.4 W (well within the 15 W TDP ceiling), a memory footprint of 2.3 GB / 8.0 GB LPDDR5, and an operating temperature of 54.2°C. Ablation of the Two-Tier Failsafe engine demonstrates its criticality: relying solely on sequence AI incurs an alert trigger delay of up to 1,050 ms for abrupt microsleep events (waiting for the full T=30 window), causing a 7.8% late-alert rate. With Tier-2 Failsafe enabled, micro-geometric reflexes (<50 ms) dispatch emergency alarms in exactly 43.6 ms (reducing late alerts to 0.0%), ensuring full compliance with Euro NCAP 2026.

6. Discussion and Practical Implications

The empirical findings deliver three principal architectural insights: (1) Compressed temporal modeling over T=30 sliding windows eliminates the prohibitive latency of 3D-CNNs while resolving recurrent step-by-step delays; (2) Decoupling the central transformer from peripheral extractors provides modular future-proofing (e.g. dropping in YOLOv10 or RT-DETRv2 without retraining the temporal core); and (3) The Two-Tier Failsafe Engine and Leaky Decay algorithm provide deterministic safety boundaries and prevent intermittent driver tampering.

Hard Edge Case Mining Analysis: The 18% hard edge cases in our dataset specifically target ambiguous real-world patterns. For instance, distinguishing talking/singing from yawning: a driver talking shows high MAR momentarily, but the temporal Latch Filter (requiring MAR >= 0.45 continuously for >= 0.6s) alongside eyelid closure logic correctly avoids false alarms. Similarly, squinting due to windshield sunlight glare reduces EAR, but Gaze and Head Pose confirmation verifies the driver is actively looking at the road, preventing false microsleep alarms.

Compliance with Euro NCAP 2026: Euro NCAP mandates warning latencies <1.5s for microsleep and <2.0s for long distraction. Operating at 45.8 FPS (21.8 ms/frame), GuardCabin DMS flags Level 3 microsleep in exactly 1.15s (35 frames) and Level 2 distraction in 1.5s (45 frames), providing comfortable compliance margins.

Automotive CAN-FD Bus and Haptic Integration: For vehicular production deployment, GuardCabin DMS interfaces with the vehicle CAN-FD network (ISO 11898-1:2015). Safety status packets are broadcast on arbitration ID 0x18F00503 every 20 ms, conveying driver focus scores, alert tiers (L0-L3), and sensor health diagnostics. Level 3 red alerts simultaneously trigger active haptic steering wheel vibration (50 Hz pulse) and acoustic buzzer overrides via CAN commands, eliminating cabin visual distraction risks.

7. Conclusion and Future Directions

We developed GuardCabin DMS, a closed-loop real-time multimodal driver monitoring system complying with Euro NCAP 2026 and ISO 22839 standards. Evaluated on 2,400 balanced sequences with 18% hard edge cases, the system achieves 97.56% accuracy, 97.57% Macro F1, 1.07 ms sequence latency, and 45.8 FPS on an edge Jetson Orin Nano.

Future Directions: (1) Unsupervised Anomaly Detection using Deep Autoencoders on the 256-D fused feature space to flag novel acute medical crises (syncope, seizures) without manual labels; (2) INT8 TinyML quantization (<3MB model footprint for <5W automotive ECUs); and (3) Active 940nm NIR illumination with CAN Bus and haptic seat/wheel feedback.

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